

## Scala For DS

### Practical no:-14

|                   |                                      |
|-------------------|--------------------------------------|
| Roll No. : S019   | Name: SUMEET KUMAVAT                 |
| Class : SY BSC IT | Date/Time of Submission : 11-08-2026 |

**Aim:- Perform basic time series analysis in Scala. Generate synthetic time series data (e.g., daily sales over a month).**

#### **CODE:-**

```
import scala.util.Random

object Practical14 extends App { new *
  println("Day\tSales")
  for (day <- 1 to 30) {
    val sales = 50 + day * 2 + Random.nextInt(10) - 5
    println(s"$day\t$sales")
  }
  println("\nsumeet -----S019")
}
```

## OUTPUT:-

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
```

```
Day Sales
```

```
1 51
```

```
2 53
```

```
3 60
```

```
4 58
```

```
5 56
```

```
6 57
```

```
7 63
```

```
8 69
```

```
9 68
```

```
10 74
```

```
11 71
```

```
12 74
```

```
13 72
```

```
14 76
```

```
15 84
```

```
16 83
```

```
17 80
```

```
18 90
```

```
19 83
```

```
20 90
```

```
21 90
```

```
22 91
```

```
23 99
```

```
24 97
```

```
25 97
```

```
26 102
```

```
27 100
```

```
28 110
```

```
29 106
```

```
30 108
```

```
sumeet ____S019
```

```
Process finished with exit code 0
```

## Scala For DS

### Practical no:-15

|                   |                                      |
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**Aim:-** Create polynomial features from a dataset. Given a list of numbers (e.g., [1, 2, 3]), generate polynomial features up to degree 3.

#### **CODE:-**

```
object Practical14 extends App { new *  
    
  val data=List(2, 3, 4, 5, 6)  
  for (x <- data) {  
    val p1 = math.pow(x, 1)  
    val p2 = math.pow(x, 2)  
    val p3 = math.pow(x, 3)  
    println(s"$x\t$p1, \t$p2, \t$p3")  
  }  
  println("\nsumeet _____S019")  
}
```

#### **OUTPUT:-**

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...  
2  2.0,    4.0,    8.0  
3  3.0,    9.0,   27.0  
4  4.0,   16.0,   64.0  
5  5.0,   25.0,  125.0  
6  6.0,   36.0,  216.0  
  
sumeet _____S019  
  
Process finished with exit code 0  
|
```

## Scala For DS

### Practical no:-16

|                   |                                      |
|-------------------|--------------------------------------|
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**Aim:-** Create a scatter plot of random data using Breeze-viz. Label the axes and customize the color of points.

**CODE:-**

```
import breeze.plot._
import scala.util.Random
object Practical14 extends App { new {

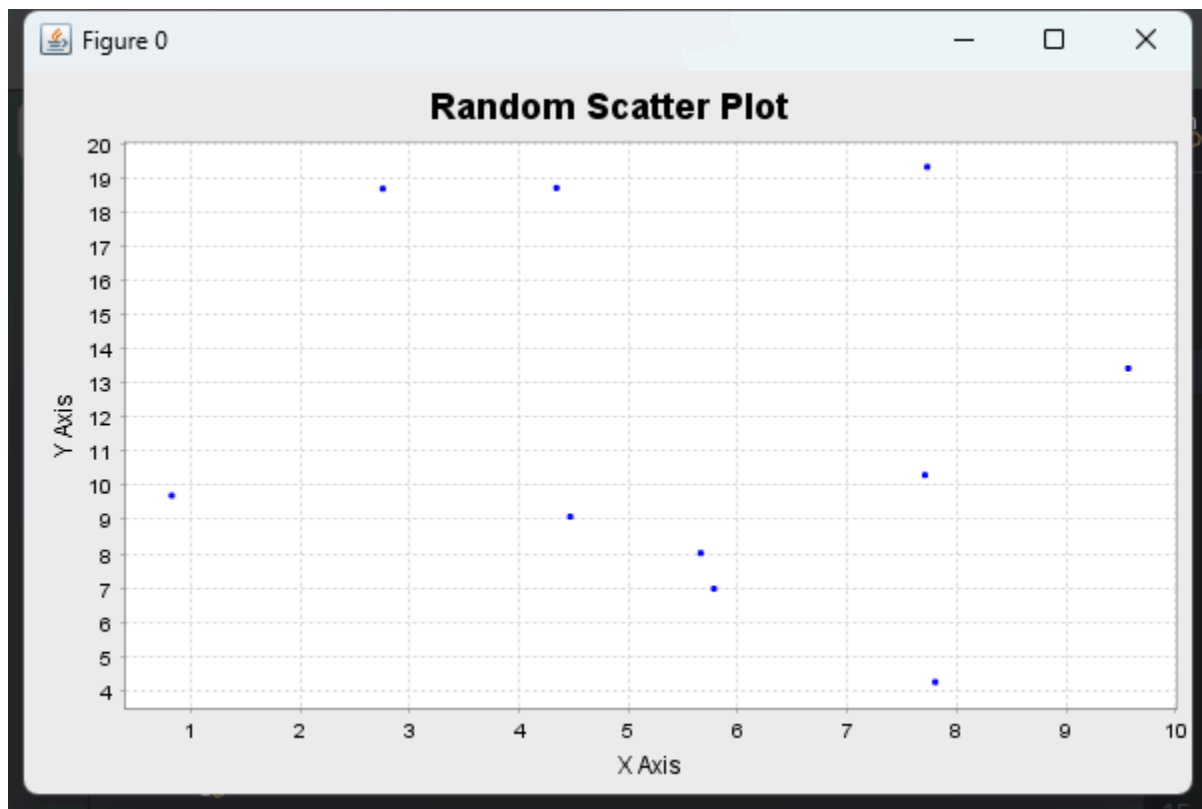
  val x = List.fill(10)(Random.nextDouble() * 10)
  val y = List.fill(10)(Random.nextDouble() * 20)
  println(x)
  println(y)
  val f = Figure()
  val p = f.subplot(0)
  p += plot(x, y, '.', colorcode = "blue")
  p.xlabel = "X Axis"
  p.ylabel = "Y Axis"
  p.title = "Random Scatter Plot"
  f.refresh()

  println("\nsumeet ____S019")
}
```

**OUTPUT:-**

```
"C:\Program Files\Java\jdk1.8.0_101\bin\java.exe" ...
List(4.341448885722356, 2.755663872937919, 7.729819117470278, 9.566045813277839, 7.708649164549987, 5.660069247102996, 0.8270961769415897, 4.467139504537988, 7.803304489208259, 5.78124
List(18.69386167635648, 18.671208862430586, 19.30941857708343, 13.421197134962693, 10.300643903989577, 8.021299816447119, 9.705815343954061, 9.08546102910021, 4.25084489427258, 6.98135
sumeet ____S019
```

## Random Scatter Plot:-



## Scala For DS

### Practical no:-17

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**AIM:- Generate a histogram of a dataset using breeze-viz. Experiment with different bin sizes.**

#### **CODE:-**

```
import breeze.plot._
import breeze.linalg.DenseVector
import scala.io.Source

object Practical14 extends App {
  val file = Source.fromResource("iris.data.csv")

  val dataList: List[Double] = file.getLines()
    .drop(1)
    .filter(_.trim.nonEmpty)
    .map(line => line.split(",")(0).trim.toDouble)
    .toList

  file.close()

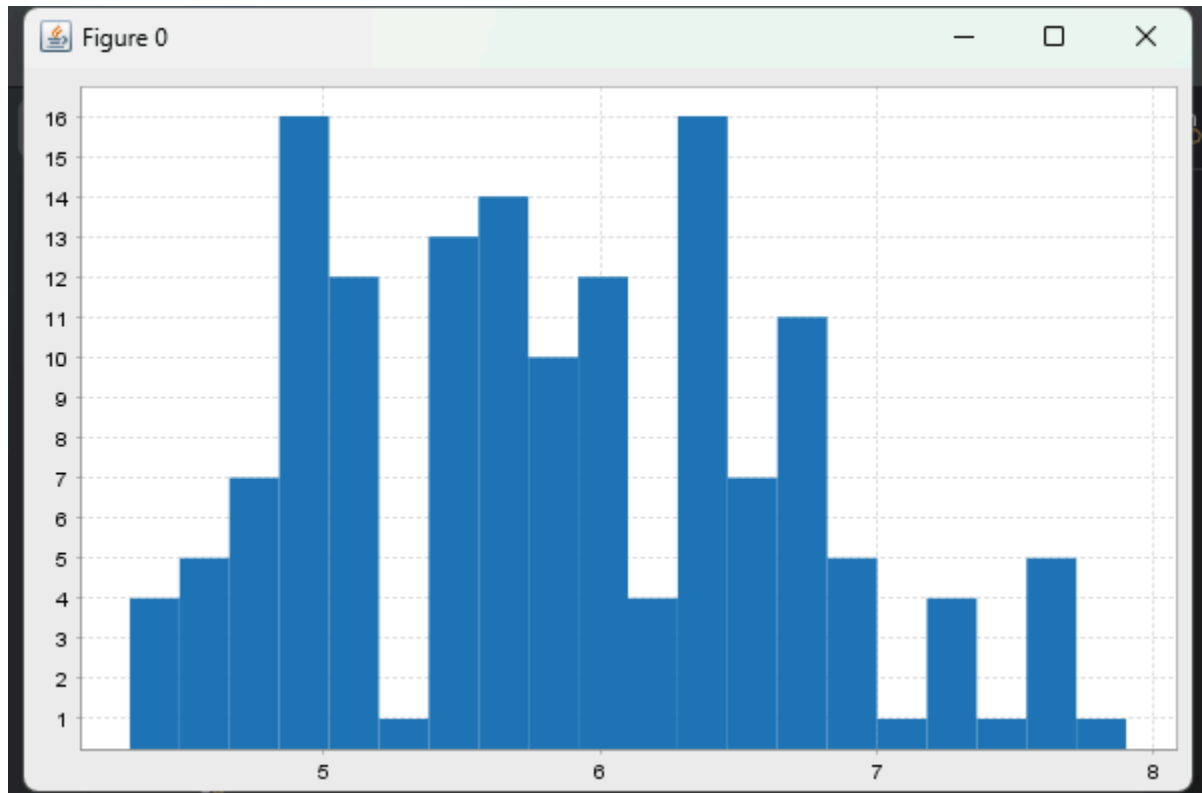
  val data = DenseVector(dataList.toArray)

  val fig = Figure()
  val plt = fig.subplot(0)

  plt += hist(data, bins = 20)

  fig.refresh()
}
```

**OUTPUT:-**



**Scala For DS**

**Practical no:-18**

|                   |                                      |
|-------------------|--------------------------------------|
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**AIM:-**plot a line graph for a dataset showing a trend over time.

**CODE:-**

```
import breeze.plot._
import breeze.linalg.DenseVector
import scala.io.Source

object Practical14 extends App {

    val file = Source.fromResource("iris.data.csv")

    // Filter out empty lines and parse Sepal Length values
    val xData: List[Double] = file.getLines()
        .drop(1)
        .filter(_.trim.nonEmpty)
        .map(line => line.split(",")(0).trim.toDouble)
        .toList

    file.close()

    // Convert to Breeze DenseVectors for reliable plotting
    val x = DenseVector(xData.toArray)
    val y = DenseVector.tabulate(x.length)(_.toDouble)

    val fig = Figure()
    val p = fig.subplot(0)

    p += plot(y, x)

    p.xlabel = "Record Number"
    p.ylabel = "Sepal Length"
    p.title = "Sepal Length Trend"

    fig.refresh()
```



OUTPUT:-

